

# Hossein Vahidi

Doctoral Researcher · Aalto University  
Department of Computer Science

Apr 2022–present

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## Research interests

I am broadly interested in the theoretical foundations of distributed and parallel computing, computational complexity, and algorithm design, particularly for graph problems. Recently, I have been studying primitives such as matrix multiplication in parallel computing models, including MPC, Congested Clique, NCC, and low-bandwidth models.

## Education

**Ph.D., Computer Science** — Aalto University Jun 2022–present  
Expected 2027; Supervisor: Prof. Jukka Suomela.

**M.Sc., Computer Science** — Saarland University Oct 2018–Oct 2021  
Thesis: Approximate Minimum Directed Spanning Tree in Congest and Congested Clique Models.  
Supervisor: Prof. Christoph Lenzen.

**B.Sc., Computer Engineering** — Shahid Bahonar University 2014–2018  
GPA: 18.67/20; Ranked 1st.  
Thesis: FIM, a Flow based framework for solving Influence Maximization Problem.  
Supervisor: Dr. Hossein Ghasemalizadeh.

## Work experience

**Doctoral Researcher** — Aalto University Apr 2022–present  
Distributed Algorithms Group · Department of Computer Science

**IT Administrator** — Max Planck Institute for Informatics Aug 2020–Mar 2022  
Department 1: Algorithms and Complexity

**Research Intern** — Aalto University Aug 2018–Sep 2018  
Distributed Algorithms Group  
Distributed recoloring with Jukka Suomela.

**Research Intern** — Max Planck Institute for Informatics Jul 2018  
Theory of Distributed and Embedded Systems Group · D1  
Anti-Ramsey coloring with Christoph Lenzen and Saeed Amiri.

## Publications

Authors are listed alphabetically by surname, following the convention in the theoretical computer science community.

- [1] **Rectangular Matrix Multiplication in the Low-Bandwidth Model**  
Chetan Gupta, Jukka Suomela, and Hossein Vahidi  
Preprint. 2026. [arXiv:2606.04652](#). [[arXiv](#)]
- [2] **Complexity of computing the anti-Ramsey numbers for paths**  
Saeed Akhoondian Amiri, Alexandru Popa, Mohammad Roghani, Golnoosh Shahkarami, Reza Soltani, and Hossein Vahidi  
Journal article. Theoretical Computer Science 1044, 2025, article 115271. [[arXiv](#)] [[Journal version](#)] [[Conference version](#)]
- [3] **Low-Bandwidth Matrix Multiplication: Faster Algorithms and More General Forms of Sparsity**  
Chetan Gupta, Janne H. Korhonen, Jan Studený, Jukka Suomela, and Hossein Vahidi  
Conference paper. SIROCCO 2025, pp. 333–349. [[arXiv](#)] [[Conference version](#)]

- [4] **Brief Announcement: Low-Bandwidth Matrix Multiplication: Faster Algorithms and More General Forms of Sparsity**  
Chetan Gupta, Janne H. Korhonen, Jan Studený, Jukka Suomela, and **Hossein Vahidi**  
Brief announcement. SPAA 2024, pp. 305–307. [arXiv \[full paper\]](#) [[Conference version](#)]
- [5] **Fast Dynamic Programming in Trees in the MPC Model**  
Chetan Gupta, Rustam Latypov, Yannic Maus, Shreyas Pai, Simo Särkkä, Jan Studený, Jukka Suomela, Jara Uitto, and **Hossein Vahidi**  
Conference paper. SPAA 2023, pp. 443–453. [[arXiv](#)] [[Conference version](#)]
- [6] **Approximate Minimum Directed Spanning Trees Under Congestion**  
Christoph Lenzen and **Hossein Vahidi**  
Conference paper. SIROCCO 2021, pp. 352–369. [[Conference version](#)]
- [7] **Complexity of Computing the Anti-Ramsey Numbers for Paths**  
Saeed Akhoondian Amiri, Alexandru Popa, Mohammad Roghani, Golnoosh Shahkarami, Reza Soltani, and **Hossein Vahidi**  
Conference paper. MFCS 2020, pp. 6:1–6:14. [[arXiv](#)] [[Conference version](#)] [[Journal version](#)]

## Academic service

Reviewer: DISC (2023, 2026); OPODIS (2026); PODC (2023); SIROCCO (2021).  
Local organizing team member: ALGO 2018.

## Teaching

### Teaching assistant, Aalto University

|                                      |            |
|--------------------------------------|------------|
| Competitive Programming              | 2024       |
| Programming Parallel Computers       | 2022, 2023 |
| Distributed Algorithms               | 2023       |
| Principles of Algorithmic Techniques | 2022       |

## Thesis advising

**Master’s thesis advisor** — Tianxing Wu, Aalto University 2025  
Towards the Evaluation of Congested Clique Compatibility on Modern Computers  
Supervisor: Prof. Jukka Suomela.

## Awards and distinctions

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| Scholarship, Saarbrücken Graduate School of Computer Science.                                            | 2018–2020 |
| Afzalipour Award for Outstanding B.Sc. Student, Shahid Bahonar University of Kerman (university-wide).   | May 2018  |
| Iranian national M.Sc. entrance examinations: 4th in Computer Science; 16th in Software Engineering.     | 2018      |
| 15th in the national final, 23rd Iranian National University Students’ Olympiad in Computer Engineering. | Aug 2017  |
| 6th place (team), ACM-ICPC Asia Tehran Regional Contest.                                                 | Dec 2015  |

## Skills

**Programming.** C, C++, Python — algorithm implementation.

**Parallel computing.** OpenMP and CUDA; SIMD vectorization; CPU memory/cache and GPU memory optimization; benchmarking and performance analysis; MPI programming.

**Formalization.** Lean 4 (working knowledge); formalized distributed and parallel computing results.

**Tools.** Git; LaTeX.

## Languages

English (fluent); Persian (native).